

Thrombocytopenia in Wiskott-Aldrich Syndrome (WAS)

The thrombocytopenia in Wiskott-Aldrich Syndrome (WAS) is persistent, with platelet counts typically ranging from 1,000 to 80,000 platelets per μL . A platelet count of $<70,000/\mu\text{L}$ is required to meet formal diagnostic criteria. The platelets are characteristically small in size, and platelet aggregation is defective.

Immunoglobulin levels are abnormal: IgM and sometimes IgG are low, and isohemagglutinins are absent. In contrast, IgA, IgE, and IgD levels are usually elevated. Hematologic abnormalities such as eosinophilia, leukopenia, and lymphopenia are commonly observed. Delayed hypersensitivity skin-test reactivity is diminished, and patients fail to respond to polysaccharide antigens.

Prognosis, Clinical Course, and Treatment

With modern therapeutic interventions, some patients with WAS survive into adulthood; however, a significant proportion die before age 10 due to hemorrhage, infection, malignancy, or complications of transplantation. Approximately 13% of patients develop lymphoreticular malignancies, particularly non-Hodgkin lymphoma, often with extranodal and central nervous system involvement. The development of autoimmune hemolytic anemia is a poor prognostic indicator and is associated with an increased risk of lymphoid malignancies; overall, 25% of patients with autoimmunity go on to develop malignancy. About 10% of patients die from these malignancies, typically during adolescence or early adulthood.

Female carriers of WAS mutations can be identified by selective inactivation of the abnormal X chromosome in T cells, B cells, and platelets. Analysis of maternal X-chromosome inactivation patterns has also been used to diagnose atypical cases in males. Mutation analysis enables prenatal diagnosis.

Management includes appropriate use of antibiotics, immunizations, and transfusions of platelets and plasma to reduce the risk of fatal infections and hemorrhage. Intravenous γ -globulin infusions are beneficial in some patients. Splenectomy may be considered for severe, recurrent bleeding, but it increases the risk of infection from encapsulated bacteria, especially in patients unable to respond to vaccines.

Bone marrow or stem cell transplantation is the treatment of choice for patients with recurrent complications, particularly significant autoimmunity. Successful engraftment leads to normalization of platelet count and function, restoration of immune function, and resolution of dermatitis (dependent on T lymphocyte engraftment). The 5-year survival rate with HLA-matched sibling donors is 87%.

Topical glucocorticoids and systemic γ -globulin may improve dermatitis. Chronic oral acyclovir is recommended for patients with recurrent eczema herpeticum.

Severe Combined Immunodeficiency (SCID)

At a Glance

- Heterogeneous group of X-linked and autosomal recessive disorders characterized by defective cell-mediated and humoral immunity.
- Clinical features: failure to thrive in infancy, chronic diarrhea, recurrent mucocutaneous candidiasis, and severe bacterial and viral infections.
- High risk of graft-versus-host disease.
- Hematopoietic stem cell transplantation can prevent early death.

Epidemiology

Severe combined immunodeficiency (SCID) comprises a group of heterogeneous disorders with similar clinical and immunologic features—profound deficiencies in both humoral and cell-mediated immunity. The overall incidence is approximately 1 in 75,000 live births. Inheritance patterns include X-linked and autosomal recessive forms.

Pathogenesis

All patients with SCID share core clinical features and exhibit defects in both T-cell and B-cell immunity, though the severity varies. The molecular basis of SCID involves disruptions in critical steps of T-cell development (see Table 144-4). A specific genetic defect is identified in about 95% of cases.

X-linked SCID accounts for approximately 46% of cases and results from mutations in the gene encoding the γ_c chain of the IL-2 receptor. This γ_c chain is a shared component of seven cytokine receptors (including those for IL-4, IL-7, IL-9, IL-15, and IL-21), explaining the profound impact on lymphocyte development and function.

Because these cytokine signals are essential for early T-cell development, female carriers exhibit skewed X-chromosome inactivation: T cells expressing the mutated allele fail to survive, leading to selective advantage of T cells with the normal allele.